

实验2 Bomb_lab

1.准备好相应文件及环境

```
racalia202212063020@LAPTO  ×  +  ∨  
racalia202212063020@LAPTOP-4F56E6J4:~$ sudo apt-get install gdb  
[sudo] password for racalia202212063020:  
Sorry, try again.  
[sudo] password for racalia202212063020:  
Sorry, try again.  
[sudo] password for racalia202212063020:  
Reading package lists... Done  
Building dependency tree  
Reading state information... Done  
gdb is already the newest version (9.2-0ubuntu1~20.04.1).  
0 upgraded, 0 newly installed, 0 to remove and 111 not upgraded.  
racalia202212063020@LAPTOP-4F56E6J4:~$ cd CUC_CSAPP  
racalia202212063020@LAPTOP-4F56E6J4:~/CUC_CSAPP$ cd \#2_Bomblab/  
racalia202212063020@LAPTOP-4F56E6J4:~/CUC_CSAPP/#2_Bomblab$ ls  
README  bomb  bomb.c  
racalia202212063020@LAPTOP-4F56E6J4:~/CUC_CSAPP/#2_Bomblab$ |
```

2.准备bomb.c的反汇编文件

```
racalia202212063020@THESELECTED3:~/CUC_CSAPP/#2_Bomblab$ objdump -d bomb > bomb.s  
racalia202212063020@THESELECTED3:~/CUC_CSAPP/#2_Bomblab$ ls  
README  bomb  bomb.c  bomb.s  
racalia202212063020@THESELECTED3:~/CUC_CSAPP/#2_Bomblab$ |
```

3.打开bomb.s并查看每个阶段的反编译数据

```
0000000000400ee0 <phase_1>:
400ee0:      48 83 ec 08      sub    $0x8,%rsp
400ee4:      be 00 24 40 00   mov    $0x402400,%esi
400ee9:      e8 4a 04 00 00   callq 401338 <strings_not_equal>
400eee:      85 c0             test   %eax,%eax
400ef0:      74 05             je     400ef7 <phase_1+0x17>
400ef2:      e8 43 05 00 00   callq 40143a <explode_bomb>
400ef7:      48 83 c4 08      add    $0x8,%rsp
400efb:      c3              retq
```

```
0000000000400efc <phase_2>:
400efc:      55              push   %rbp
400efd:      53              push   %rbx
400efe:      48 83 ec 28     sub    $0x28,%rsp
400f02:      48 89 e6        mov    %rsp,%rsi
400f05:      e8 52 05 00 00   callq 40145c <read_six_numbers>
400f0a:      83 3c 24 01     cmpl   $0x1,(%rsp)
400f0e:      74 20             je     400f30 <phase_2+0x34>
400f10:      e8 25 05 00 00   callq 40143a <explode_bomb>
400f15:      eb 19             jmp    400f30 <phase_2+0x34>
400f17:      8b 43 fc        mov    -0x4(%rbx),%eax
400f1a:      01 c0             add    %eax,%eax
400f1c:      39 03             cmp    %eax,(%rbx)
400f1e:      74 05             je     400f25 <phase_2+0x29>
400f20:      e8 15 05 00 00   callq 40143a <explode_bomb>
400f25:      48 83 c3 04     add    $0x4,%rbx
400f29:      48 39 eb        cmp    %rbp,%rbx
400f2c:      75 e9             jne    400f17 <phase_2+0x1b>
400f2e:      eb 0c             jmp    400f3c <phase_2+0x40>
400f30:      48 8d 5c 24 04   lea    0x4(%rsp),%rbx
400f35:      48 8d 6c 24 18   lea    0x18(%rsp),%rbp
400f3a:      eb db             jmp    400f17 <phase_2+0x1b>
400f3c:      48 83 c4 28     add    $0x28,%rsp
400f40:      5b              pop     %rbx
400f41:      5d              pop     %rbp
400f42:      c3              retq
```

```

0000000000400f43 <phase_3>:
 400f43: 48 83 ec 18      sub    $0x18,%rsp
 400f47: 48 8d 4c 24 0c    lea    0xc(%rsp),%rcx
 400f4c: 48 8d 54 24 08    lea    0x8(%rsp),%rdx
 400f51: be cf 25 40 00    mov    $0x4025cf,%esi
 400f56: b8 00 00 00 00    mov    $0x0,%eax
 400f5b: e8 90 fc ff ff    callq 400bf0 <__isoc99_sscanf@plt>
 400f60: 83 f8 01          cmp    $0x1,%eax
 400f63: 7f 05            jg     400f6a <phase_3+0x27>
 400f65: e8 d0 04 00 00    callq 40143a <explode_bomb>
 400f6a: 83 7c 24 08 07    cmpl   $0x7,0x8(%rsp)
 400f6f: 77 3c            ja     400fad <phase_3+0x6a>
 400f71: 8b 44 24 08      mov    0x8(%rsp),%eax
 400f75: ff 24 c5 70 24 40 00 jmpq    *0x402470(,%rax,8)
 400f7c: b8 cf 00 00 00    mov    $0xcf,%eax
 400f81: eb 3b            jmp     400fbe <phase_3+0x7b>
 400f83: b8 c3 02 00 00    mov    $0x2c3,%eax
 400f88: eb 34            jmp     400fbe <phase_3+0x7b>
 400f8a: b8 00 01 00 00    mov    $0x100,%eax
 400f8f: eb 2d            jmp     400fbe <phase_3+0x7b>
 400f91: b8 85 01 00 00    mov    $0x185,%eax
 400f96: eb 26            jmp     400fbe <phase_3+0x7b>
 400f98: b8 ce 00 00 00    mov    $0xce,%eax
 400f9d: eb 1f            jmp     400fbe <phase_3+0x7b>
 400f9f: b8 aa 02 00 00    mov    $0x2aa,%eax
 400fa4: eb 18            jmp     400fbe <phase_3+0x7b>
 400fa6: b8 47 01 00 00    mov    $0x147,%eax
 400fab: eb 11            jmp     400fbe <phase_3+0x7b>
 400fad: e8 88 04 00 00    callq 40143a <explode_bomb>
 400fb2: b8 00 00 00 00    mov    $0x0,%eax
 400fb7: eb 05            jmp     400fbe <phase_3+0x7b>
 400fb9: b8 37 01 00 00    mov    $0x137,%eax
 400fbe: 3b 44 24 0c      cmp    0xc(%rsp),%eax
 400fc2: 74 05            je     400fc9 <phase_3+0x86>
 400fc4: e8 71 04 00 00    callq 40143a <explode_bomb>
 400fc9: 48 83 c4 18      add    $0x18,%rsp
 400fcd: c3              retq

```

```

000000000040100c <phase_4>:
 40100c: 48 83 ec 18      sub    $0x18,%rsp
 401010: 48 8d 4c 24 0c    lea    0xc(%rsp),%rcx
 401015: 48 8d 54 24 08    lea    0x8(%rsp),%rdx
 40101a: be cf 25 40 00    mov    $0x4025cf,%esi
 40101f: b8 00 00 00 00    mov    $0x0,%eax
 401024: e8 c7 fb ff ff    callq  400bf0 <__isoc99_sscanf@plt>
 401029: 83 f8 02          cmp     $0x2,%eax
 40102c: 75 07            jne     401035 <phase_4+0x29>
 40102e: 83 7c 24 08 0e    cmpl    $0xe,0x8(%rsp)
 401033: 76 05            jbe     40103a <phase_4+0x2e>
 401035: e8 00 04 00 00    callq  40143a <explode_bomb>
 40103a: ba 0e 00 00 00    mov     $0xe,%edx
 40103f: be 00 00 00 00    mov     $0x0,%esi
 401044: 8b 7c 24 08      mov     0x8(%rsp),%edi
 401048: e8 81 ff ff ff    callq  400fce <func4>
 40104d: 85 c0            test    %eax,%eax
 40104f: 75 07            jne     401058 <phase_4+0x4c>
 401051: 83 7c 24 0c 00    cmpl    $0x0,0xc(%rsp)
 401056: 74 05            je      40105d <phase_4+0x51>
 401058: e8 dd 03 00 00    callq  40143a <explode_bomb>
 40105d: 48 83 c4 18      add     $0x18,%rsp
 401061: c3              retq

```

```

0000000000401062 <phase_5>:
 401062:      53                                push    %rbx
 401063:      48 83 ec 20                      sub     $0x20,%rsp
 401067:      48 89 fb                          mov     %rdi,%rbx
 40106a:      64 48 8b 04 25 28 00            mov     %fs:0x28,%rax
 401071:      00 00
 401073:      48 89 44 24 18                  mov     %rax,0x18(%rsp)
 401078:      31 c0                            xor     %eax,%eax
 40107a:      e8 9c 02 00 00                  callq   40131b <string_length>
 40107f:      83 f8 06                          cmp     $0x6,%eax
 401082:      74 4e                            je      4010d2 <phase_5+0x70>
 401084:      e8 b1 03 00 00                  callq   40143a <explode_bomb>
 401089:      eb 47                            jmp     4010d2 <phase_5+0x70>
 40108b:      0f b6 0c 03                      movzbl  (%rbx,%rax,1),%ecx
 40108f:      88 0c 24                          mov     %cl,(%rsp)
 401092:      48 8b 14 24                      mov     (%rsp),%rdx
 401096:      83 e2 0f                          and     $0xf,%edx
 401099:      0f b6 92 b0 24 40 00            movzbl  0x4024b0(%rdx),%edx
 4010a0:      88 54 04 10                      mov     %dl,0x10(%rsp,%rax,1)
 4010a4:      48 83 c0 01                      add     $0x1,%rax
 4010a8:      48 83 f8 06                      cmp     $0x6,%rax
 4010ac:      75 dd                            jne     40108b <phase_5+0x29>
 4010ae:      c6 44 24 16 00                  movb    $0x0,0x16(%rsp)
 4010b3:      be 5e 24 40 00                  mov     $0x40245e,%esi
 4010b8:      48 8d 7c 24 10                  lea     0x10(%rsp),%rdi
 4010bd:      e8 76 02 00 00                  callq   401338 <strings_not_equal>
 4010c2:      85 c0                            test    %eax,%eax
 4010c4:      74 13                            je      4010d9 <phase_5+0x77>
 4010c6:      e8 6f 03 00 00                  callq   40143a <explode_bomb>
 4010cb:      0f 1f 44 00 00                  nopl    0x0(%rax,%rax,1)
 4010d0:      eb 07                            jmp     4010d9 <phase_5+0x77>
 4010d2:      b8 00 00 00 00                  mov     $0x0,%eax
 4010d7:      eb b2                            jmp     40108b <phase_5+0x29>
 4010d9:      48 8b 44 24 18                  mov     0x18(%rsp),%rax
 4010de:      64 48 33 04 25 28 00            xor     %fs:0x28,%rax
 4010e5:      00 00
 4010e7:      74 05                            je      4010ee <phase_5+0x8c>
 4010e9:      e8 42 fa ff ff                  callq   400b30 <__stack_chk_fail@plt>
 4010ee:      48 83 c4 20                      add     $0x20,%rsp
 4010f2:      5b                                pop     %rbx
 4010f3:      c3                                retq

```

00000000004010f4 <phase_6>:

4010f4:	41 56	push	%r14
4010f6:	41 55	push	%r13
4010f8:	41 54	push	%r12
4010fa:	55	push	%rbp
4010fb:	53	push	%rbx
4010fc:	48 83 ec 50	sub	\$0x50,%rsp
401100:	49 89 e5	mov	%rsp,%r13
401103:	48 89 e6	mov	%rsp,%rsi
401106:	e8 51 03 00 00	callq	40145c <read_six_numbers>
40110b:	49 89 e6	mov	%rsp,%r14
40110e:	41 bc 00 00 00 00	mov	\$0x0,%r12d
401114:	4c 89 ed	mov	%r13,%rbp
401117:	41 8b 45 00	mov	0x0(%r13),%eax
40111b:	83 e8 01	sub	\$0x1,%eax
40111e:	83 f8 05	cmp	\$0x5,%eax
401121:	76 05	jbe	401128 <phase_6+0x34>
401123:	e8 12 03 00 00	callq	40143a <explode_bomb>
401128:	41 83 c4 01	add	\$0x1,%r12d
40112c:	41 83 fc 06	cmp	\$0x6,%r12d
401130:	74 21	je	401153 <phase_6+0x5f>
401132:	44 89 e3	mov	%r12d,%ebx
401135:	48 63 c3	movslq	%ebx,%rax
401138:	8b 04 84	mov	(%rsp,%rax,4),%eax
40113b:	39 45 00	cmp	%eax,0x0(%rbp)
40113e:	75 05	jne	401145 <phase_6+0x51>
401140:	e8 f5 02 00 00	callq	40143a <explode_bomb>
401145:	83 c3 01	add	\$0x1,%ebx
401148:	83 fb 05	cmp	\$0x5,%ebx
40114b:	7e e8	jle	401135 <phase_6+0x41>
40114d:	49 83 c5 04	add	\$0x4,%r13
401151:	eb c1	jmp	401114 <phase_6+0x20>
401153:	48 8d 74 24 18	lea	0x18(%rsp),%rsi
401158:	4c 89 f0	mov	%r14,%rax
40115b:	b9 07 00 00 00	mov	\$0x7,%ecx
401160:	89 ca	mov	%ecx,%edx
401162:	2b 10	sub	(%rax),%edx
401164:	89 10	mov	%edx,(%rax)
401166:	48 83 c0 04	add	\$0x4,%rax
40116a:	48 39 f0	cmp	%rsi,%rax
40116d:	75 f1	jne	401160 <phase_6+0x6c>
40116f:	be 00 00 00 00	mov	\$0x0,%esi
401174:	eb 21	jmp	401197 <phase_6+0xa3>

401176:	48 8b 52 08	mov	0x8(%rdx),%rdx
40117a:	83 c0 01	add	\$0x1,%eax
40117d:	39 c8	cmp	%ecx,%eax
40117f:	75 f5	jne	401176 <phase_6+0x82>
401181:	eb 05	jmp	401188 <phase_6+0x94>
401183:	ba d0 32 60 00	mov	\$0x6032d0,%edx
401188:	48 89 54 74 20	mov	%rdx,0x20(%rsp,%rsi,2)
40118d:	48 83 c6 04	add	\$0x4,%rsi
401191:	48 83 fe 18	cmp	\$0x18,%rsi
401195:	74 14	je	4011ab <phase_6+0xb7>
401197:	8b 0c 34	mov	(%rsp,%rsi,1),%ecx
40119a:	83 f9 01	cmp	\$0x1,%ecx
40119d:	7e e4	jle	401183 <phase_6+0x8f>
40119f:	b8 01 00 00 00	mov	\$0x1,%eax
4011a4:	ba d0 32 60 00	mov	\$0x6032d0,%edx
4011a9:	eb cb	jmp	401176 <phase_6+0x82>
4011ab:	48 8b 5c 24 20	mov	0x20(%rsp),%rbx
4011b0:	48 8d 44 24 28	lea	0x28(%rsp),%rax
4011b5:	48 8d 74 24 50	lea	0x50(%rsp),%rsi
4011ba:	48 89 d9	mov	%rbx,%rcx
4011bd:	48 8b 10	mov	(%rax),%rdx
4011c0:	48 89 51 08	mov	%rdx,0x8(%rcx)
4011c4:	48 83 c0 08	add	\$0x8,%rax
4011c8:	48 39 f0	cmp	%rsi,%rax
4011cb:	74 05	je	4011d2 <phase_6+0xde>
4011cd:	48 89 d1	mov	%rdx,%rcx
4011d0:	eb eb	jmp	4011bd <phase_6+0xc9>
4011d2:	48 c7 42 08 00 00 00	movq	\$0x0,0x8(%rdx)
4011d9:	00		
4011da:	bd 05 00 00 00	mov	\$0x5,%ebp
4011df:	48 8b 43 08	mov	0x8(%rbx),%rax
4011e3:	8b 00	mov	(%rax),%eax
4011e5:	39 03	cmp	%eax,(%rbx)
4011e7:	7d 05	jge	4011ee <phase_6+0xfa>
4011e9:	e8 4c 02 00 00	callq	40143a <explode_bomb>
4011ee:	48 8b 5b 08	mov	0x8(%rbx),%rbx
4011f2:	83 ed 01	sub	\$0x1,%ebp
4011f5:	75 e8	jne	4011df <phase_6+0xeb>
4011f7:	48 83 c4 50	add	\$0x50,%rsp
4011fb:	5b	pop	%rbx
4011fc:	5d	pop	%rbp
4011fd:	41 5c	pop	%r12
4011ff:	41 5d	pop	%r13

4.拆弹准备

1.phase1根据反编译代码中的代码print出正确的密码

```
Reading symbols from bomb...
(gdb) print (char*)0x402400
$1 = 0x402400 "Border relations with Canada have never been better."
(gdb) |
```

2.phase2根据反编译代码中的代码找到对应的数字

```
(gdb) disas read_six_numbers
Dump of assembler code for function read_six_numbers:
   0x000000000040145c <+0>:      sub     $0x18,%rsp
   0x0000000000401460 <+4>:      mov     %rsi,%rdx
   0x0000000000401463 <+7>:      lea     0x4(%rsi),%rcx
   0x0000000000401467 <+11>:     lea     0x14(%rsi),%rax
   0x000000000040146b <+15>:     mov     %rax,0x8(%rsp)
   0x0000000000401470 <+20>:     lea     0x10(%rsi),%rax
   0x0000000000401474 <+24>:     mov     %rax,(%rsp)
   0x0000000000401478 <+28>:     lea     0xc(%rsi),%r9
   0x000000000040147c <+32>:     lea     0x8(%rsi),%r8
   0x0000000000401480 <+36>:     mov     $0x4025c3,%esi
   0x0000000000401485 <+41>:     mov     $0x0,%eax
   0x000000000040148a <+46>:     callq   0x400bf0 <__isoc99_sscanf@plt>
   0x000000000040148f <+51>:     cmp     $0x5,%eax
   0x0000000000401492 <+54>:     jg      0x401499 <read_six_numbers+61>
   0x0000000000401494 <+56>:     callq   0x40143a <explode_bomb>
   0x0000000000401499 <+61>:     add     $0x18,%rsp
   0x000000000040149d <+65>:     retq
End of assembler dump.
(gdb) |
```

3.phase3找到密码的数字类型并调到相应的表，根据类似算法找出密码对应表

```
(gdb) print (char*)0x4025cf
$2 = 0x4025cf "%d %d"
```

```
(gdb) p/x *(0x402470)
$3 = 0x400f7c
(gdb) p/x *(0x402470 + 8)
$4 = 0x400fb9
(gdb) |
```

4.phase4从第5行看起，0x4025cf指向的地方存储的仍然是两个int型整数。第8行和2比较，说明输入参数的个数为2。第10行和14比较，说明输入的第一个参数一定要小于14。第13,14,15行向func4

() 传递三个参数0x8(%rsp),0,14。第17行测试函数返回值是否为0，要想不爆炸，函数返回值一定要为0。第19行说明输入的第二个参数一定要为0。所以，我们要确定的是当输入的第一个参数为多少的时候，fun4 () 的返回值为0。当x == k时，返回值为0。所以第一个参数为7

5.phase5根据反编译代码中的代码找到相应的乱码，并且找到其对应的密码表并翻译出密码

```
(gdb) x/sb 0x4024b0
0x4024b0 <array.3449>: "maduiersnfotvbylSo you think you can stop the bomb with ctrl-c
, do you?"
(gdb) x/sb 0x40245e
0x40245e: "flyers"
```

6.phase6首先，函数要求读入6个数，并确认个数是否为6。然后，通过双重循环，判断6个数之间不存在重复.这里有一个细节，就是这段代码保存了和7的差。接下来，代码要求由大到小获取链表中的

值，所以我们打印链表的节点值。从大到小排列他们的索引分别是：
3 4 5 6 1 2，考虑被 7 减的操作，所以答案是：4 3 2 1 6

5.拆除炸弹

```
(gdb) r
Starting program: /home/racalia202212063020/CUC_CSAPP/#2_Bomblab/bomb
Welcome to my fiendish little bomb. You have 6 phases with
which to blow yourself up. Have a nice day!
Border relations with Canada have never been better.
Phase 1 defused. How about the next one?
1 2 4 8 16 32
That's number 2. Keep going!
0 207
Halfway there!
7 0
So you got that one. Try this one.
IONEFG
Good work! On to the next...
4 3 2 1 6 5
Congratulations! You've defused the bomb!
[Inferior 1 (process 202) exited normally]
```